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Comparison of Modeling Methods for DoE-Based Holistic Upstream Process Characterization

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Upstream bioprocess characterization and optimization are time and resource-intensive tasks. Regularly in the biopharmaceutical industry, statistical design of experiments (DoE) in combination with response surface models (RSMs) are used, neglecting the process trajectories and dynamics. Generating process understanding with time-resolved, dynamic process models allows to understand the impact of temporal deviations, production dynamics, and provides a better understanding of the process variations that stem from the biological subsystem. The authors propose to use DoE studies in combination with hybrid modeling for process characterization. This approach is showcased on *Escherichia coli* fed-batch cultivations at the 20L scale, evaluating the impact of three critical process parameters. The performance of a hybrid model is compared to a pure data-driven model and the widely adopted RSM of the process endpoints. Further, the performance of the time-resolved models to simultaneously predict biomass and titer is evaluated. The superior behavior of the hybrid model compared to the pure black-box approaches for process characterization is presented. The evaluation considers important criteria, such as the prediction accuracy of the biomass and titer endpoints as well as the time-resolved trajectories. This showcases the high potential of hybrid models for soft-sensing and model predictive control.

1. Introduction

1.1. Quality by Design and Process Characterization

The U.S. Food and Drug Administration (FDA) proposed the process analytical technology (PAT) guidelines in 2004[1] to foster the application of science and risk-based technologies in the biopharmaceutical industry. Answering to the guidelines, industry adopts more and more a Quality by Design (QbD) paradigm to gain deeper process insights and to counteract batch-to-batch variability (e.g., fluctuations in the cultivation temperature and temporary pump malfunctions).[2] At the beginning of QbD implementation, a technical risk assessment is used to identify critical quality attributes (CQA) of the respective product and thereupon the critical process parameters (CPP) that might affect the CQAs.[3] Once the CPPs are known, process characterization studies are performed to assess and understand their impact on the CQAs. These characterization studies nowadays mostly adopt a combination of statistical design of experiments (DoE) with response surface modeling (RSM), in particular for upstream bioprocess characterization.[4–6]

1.2. Design of Experiments

The basic purpose of DoEs is to systematically investigate the relationship between defined CPPs (input factors) and the process response of interest, for example, the final product concentration. Different levels of each CPP (e.g., different pH set points or cultivation temperatures) are typically examined to have a better understanding of the respective impacts.[7] The multidimensional combination of the identified and chosen CPPs, each with a selected number of investigated levels, set up the space to be investigated. Herein, the space in which the desired product quality is achieved is referred to as the design space.[8] Depending on the specific aim of the study different designs can be applied, by performing cultivations in this space, characterizing it, and gaining process knowledge, for example, an initial screening design, investigating only the corners of the design space, or a full factorial design for process characterization and optimization. For the first use, many reduced designs can be applied, for example,

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definite screening, fractional factorial or Box-Behnken designs to name a few. All of these designs require a different number of experiments and generate different amounts of information.[9] As a result, the area of interest can be identified in which the original DoE can be extended to derive robust process settings and to describe their aftermath.[10] However, to completely characterize this space, the number of experiments suggested by DoEs can easily exceed a feasible number of experiments, if too many factors or levels are chosen or the experiment's inherent complexity is very high. This is typically accompanied by long durations of the studies and tremendous expenses.[11]

1.3. Response Surface Modeling

In process development and optimization settings, typically RSM are adopted to analyze the generated analytical results,[12] for example, using one out of the many available statistical software tools. Most often solely the impact of the factors on the endpoints of the processes is investigated.[13,14] This only leads to a snapshot of the end of the process, since the rest of the process is neglected, which can lead to significantly different conclusions than if the complete process profile would be considered. For instance, RSM are utilized to investigate the relationship between the CPPs and one or more process variables of interest, for example, the biomass and product titer.[15,16] However, when using such models for endpoint descriptions, many temporary process influences are not taken into account. Also, process dynamics are neglected, wherefore the aftermath of process deviations is not understood and the impact of changes in the CPPs during the process cannot be described.[17]

1.4. Process Modeling

More generally, there are two ways of modeling, called parametric (white box) and non-parametric (black box) modeling.[18] Non-parametric models refer to purely data-driven approaches, for which no further process knowledge is needed, the model structure is inferred from data, for example, artificial neural networks (ANN). However, once the model is used for predictions out of the characterized space, it lacks the ability to extrapolate. The structure[19] of a parametric model is based on knowledge and empirical considerations and therefore is a more suitable match for extrapolation. Since this type of model solely assumes a mechanistic trend, and does not account for empiric observations, the predictability is also often inaccurate. These parametric models are rarely used in upstream process development since their development is time consuming and laborious.[20,21] Despite their shortcomings, both approaches are already in use to model and predict processes.[22–26]

1.5. Hybrid Modeling

One way to exploit the advantages of both modeling approaches is called hybrid (semi-parametric) modeling. Hereby, it is possible to benefit from the positive aspects and make up for the respective drawbacks.[27] Such a model can either combine the black- and white-box parts in a parallel or serial structure. The difference is the order in which the different parts and their respective weights are taken into account in the overall model. These features generally make it a more cost-effective modeling approach to deal with complex problems[28]; for example, the black box can be used to estimate the rate expressions used in the white box.[29] In contrast to simple black-box endpoint models, certain hybrid model structures are able to describe the entire process and not only the endpoints. As a result, the behavior and deviations of a process can be understood and therefore it is possible to understand the impact of changing the CPPs during the process.[30]

1.6. Bridging the Research Gap

Many methods, approaches, and tools were developed since the start of the PAT initiative, to implement QbD in upstream process development, characterization, and optimization.[31] Current methods to evaluate the results obtained from DoE, for example, RSM, solely focus on the endpoint of a process, neglecting the remaining cultivation, missing the majority of the process information. The main objective of this study was to show how this gap can be closed in the biopharmaceutical industry by utilizing hybrid modeling, with respect to process understanding and compared to the performance of the most commonly adopted techniques.

We utilized experimental data from *Escherichia coli* fed-batch cultivations at the 20L scale expressing the recombinant protein human Cu/Zn superoxide dismutase (hSOD). A space with three factors, each with three levels, was characterized by performing fed-batch fermentations at each CPP combination setting, leading to 27 distinctive DoE conditions. With this generated fed-batch fermentation data, the best structure and input to the three modeling approaches were searched. Consecutively, the performance of the hybrid model was compared to the widely adopted RSM methodology and a solely data-driven ANN model.

2. Experimental Section

2.1. Process Conditions

All *E. coli* (HMS174 (DE3)) fed-batch cultivations were performed at the 20L scale expressing hSOD. The feed phase was carried out for four doubling times and induction always took place after the first doubling time, that is, the product formation persisted for the remaining three doubling times. Detailed information about the media, feed, strain, plasmid, cultivation, induction conditions, on- and off-line monitoring were already published elsewhere.[32,33] The investigated design of these studies was later extended, resulting in a design space with three CPPs, each at three levels, that is, 27 CPP combinations in total.[34] The investigated CPPs comprise the cultivation temperature (30, 34, and 37 °C), the induction strength (0.2, 0.5, and 0.9 μmol IPTG/g cell dry mass [CDM]) and the intended specific growth rate (0.10, 0.15, and 0.20 h^{-1}), which was used to compute the feeding rate, as reported in detail elsewhere.[35]

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2.2. Data Sets

The data set containing the fed-batch fermentations originated from the previous DoE study and consisted of 31 fermentations, covered the 27 CPP conditions, together with two duplicates and one triplicate. The two off-line process variables, namely the biomass concentration and the soluble product titer, were measured once prior to the induction and hourly afterward. The biomass concentration was determined using thermogravimetric analysis,[36] while for the soluble product titer, an ELISA assay was performed.[37] The on-line available process variables (every 3 minutes) comprised the standard measurements, for example, temperature, feed balance, inductor balance, base balance, stirrer speed, and inlet air. The detailed analytical results for the biomass and the soluble product titer of the fed-batch fermentations are given in Figure S1, Supporting Information. To estimate the uncertainty of the biomass and titer production process, we repeated experiments of one particular fermentation condition seven times. We calculated the errors over all samples and obtained 3.6% and 7.6% for biomass and titer accuracy, respectively. These values represent the threshold of accuracy a certain model can obtain.

2.3. Data Preprocessing

2.3.1. Standardization of the On-Line Data

Prior to the time-resolved process modeling, the on-line available process variables used for the model building were standardized along with the time domain, using the z score.

2.3.2. Interpolation of the Off-Line Data

The off-line measurements for the biomass concentration were available every hour, while the measurements for the soluble product titer were only available every 2 h. For a valid weighted evaluation, a value for the soluble product titer to each biomass value was provided by interpolation to the sampling frequency of the biomass using Hermite polynomials. For this type of interpolation, an initial value and at least four additional values of each variable (i.e., five in total) to be interpolated have to be provided.

2.4. Process Modeling

The respective fed-batch fermentations used for model building (train and validation data set) and model testing (test data set) for the different modeling approaches are described below.

1. Response surface model: all fed-batch fermentations (DoE #1-27) were used ($N = 31$).
2. ANN black-box model: 25 fed-batch fermentations (DoE #1, #3-16, #18-20, and #22 27) for model training and 6 fed-batch fermentations (DoE #2, #4, #9, #17, #21, and #22) for model testing were used ($N = 25+6$).
3. Hybrid model: the same training and test fed-batch fermentations as for the ANN black-box model were used to allow for a fair comparison ($N = 25 + 6$).

The six fed-batch fermentations in the test set were chosen in a way that one fermentation of each replicate run and three individual fed-batch fermentations, which were not present in the training and validation data set were used and that each induction strength was represented by two fed-batch fermentations. The complete list of all CPP combination settings and the respective naming is given in Table S1, Supporting Information.

The black-box model and the hybrid model were developed in the C# hybrid modeling toolbox (Novasign GmbH, Vienna, Austria).

2.4.1. Response Surface Model

The RSM of the full factorial DoE, modeling the endpoints, was generated with MATLAB (2016b, MathWorks, USA) using the function *rstool*, taking independent, constant, linear, interaction, and squared terms into account, to fit the surface and the global confidence intervals set to 95 %.

2.4.2. Artificial Neural Network Black-Box Model

An ANN, that is, the use of propagating predictions, applying a Levenberg–Marquardt algorithm was used to describe the concentration profiles of biomass and soluble product titer. Herein, the predicted values were calculated by numerical stepwise integrating the values from the previously estimated time point, as elaborated elsewhere.[38] Three input variables were used, the cultivation temperature (in °C), the cumulative inductor mass (in kg) and the cumulative feed (in L). The output variables were the biomass concentration and the soluble product titer. These model input variables were chosen due to their assumed significant impact on the soluble product titer.[33] Additionally, these parameters were good candidates to also allow the view on model predictive control (MPC) as the future perspective, since these were direct process inputs and their simple controllability. To identify the most suitable model structure, the number of hidden layers and neurons was systematically varied and the model was chosen for which the best performance in terms of the Akaike information criterion (minimal value) was obtained.

A single hidden layer of four neurons proved to be the best performing structure. Linear transfer functions for the input and output layers and hyperbolic transfer functions for the hidden layer were used.

2.4.3. Hybrid Model

A serial hybrid model was developed by complementing the knowledge-based white-box model with an ANN. Material balances were adopted to describe the evolution of biomass (Equation (1)) and product (Equation (2)). Here, the black-box part was adopted to likewise model the unknown rate expressions, μ and $v_{p/x}$:

$$\frac{dX}{dt} = \mu * X - D * X \tag{1}$$

$$\frac{dP}{dt} = \mathrm{v}_{p/x} * \mathrm{X} * I_{y/n} - \mathrm{D} * \mathrm{P} \tag{2}$$

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With the biomass concentration (X) in g L^{-1}, the soluble product titer (P) in g L^{-1}, the inductor switch ($I_{y/n}$), that is, possessing either the value 0 (no induction) or 1 (induction), and the dilution factor (D), which contains the feed addition and sampling, both in liters.

2.4.4. Model Validation

To validate the quality of the established model, an internal validation was performed. Therefore, the training data set was randomly split, fermentation wise, into a training and validation set (ratio 0.8). The training partition was used to train the black-box model and applied on the remaining validation partition. The training stopped once no further improvement in the model performance for the validation partition was achieved, that is, further model training would only lead to overfitting. To provide a large variety of different models, this partitioning of the fed-batch fermentations into boots was repeated 40 times, that is, the fed-batch fermentations used for model building were shuffled and randomly assigned to either the training or validation partition 40 times. For all established models the RMSE (Equation (3)) and the percentage model error (Equation (4)) were calculated with the measured value (y), its estimated counterpart ($\hat{y}$) for each time point (t) and the total number of observations (N).

$$\mathrm{RMSE} = \sqrt{\frac{1}{N} * \sum \left(y_{(t)} - \hat{y}_{(t)}\right)^2} \tag{3}$$

$$\mathrm{Error}\,[\%] = \frac{100}{N} * \sum \frac{\left|y_{(t)} - \hat{y}_{(t)}\right|}{y_{(t)}} \tag{4}$$

2.4.5. Bootstrapping

For the ANN black-box and the hybrid model, bootstrap aggregation was applied to enhance the robustness and evaluate its predictive uncertainty.[39] Each bootstrapped model consisted of six individual models (each derived from a different boot) to ensure that every fed-batch fermentation was present in the bootstrap-aggregated model and to guarantee a meaningful evaluation. For each bootstrapped model, the standard deviation (SD) (Equation (5)) was calculated with the bootstrapped value of the prediction ($\hat{y}_{\mathrm{btstrp}}$) (i.e., the mean value of all used models), the respective predicted counterpart from the models ($\hat{y}_{\mathrm{model}}$), the index ($i = 1{:}6$) and the number of observations for each time point (n). The SD was used to compute the prediction interval (PI) (Equation (6)). The respective PIs were derived from the SD of the incorporated models. These bootstrap-aggregated models were applied on the test set (external validation), to assess the predictability of the final models on new data.

$$\mathrm{SD}_{(t)} = \sqrt{\frac{1}{n-1} * \sum \left(\hat{y}_{\mathrm{btstrp}(t)} - \hat{y}_{\mathrm{model}(i)_{(t)}}\right)^2} \tag{5}$$

$$\mathrm{PI}_{(t)} = \hat{y}_{\mathrm{btstrp}(t)} \pm \mathrm{SD}_{(t)} \tag{6}$$

3. Results

3.1. Model Performance Comparison for Process Endpoints

We compared the model performance of the RSM, the black-box model, and the hybrid model at the process endpoints. The endpoint values of the generated black-box model and the hybrid model were taken and plotted as surfaces. This allows for a comparison to the widely adopted RSM applications, with respect to the accuracy. The response surfaces were separated by the induction strengths, referred to as induction planes. The three approaches, a commonly developed RSM using a full quadratic fitting function and the ANN black box as well as the hybrid model, were compared to the measured endpoint values. One exemplary induction plane (0.5 µmol IPTG/g CDM) of the biomass concentration (**Figure 1**) and the soluble product titer (**Figure 2**) is presented.

The summarized modeling results are listed in **Table 1**. Herein, all induction planes are incorporated for the calculations of the R^2, the RMSE, and the percentage error. Even though the same value is calculated for the RMSE, for example, for the RSM and the hybrid model, these resulted in different values in the percentage error. This is due to the relativity in the calculation method of the percentage error.

An explicit trend toward higher biomass concentrations with CPP combinations using faster specific growth rates, lower induction strengths, and lower temperatures was observed in all induction planes. Compared to the experimental response for the biomass concentration (Figure 1A), the RSM exhibits high similarity (Figure 1B). The response surface of the black-box model (Figure 1C) is of limited quality and does not allow for meaningful comparison, for example, displaying a declining biomass concentration in the region, where the highest biomass concentration was observed, accompanied by high SDs. Two of the replicates in the test set were performed in this induction plane, respectively, fed-batch fermentation #4 (30 °C and μ 0.10) and #22 (30 °C and μ 0.20). Even though these data sets were present in the training and the test set, the estimation from the black-box model was error prone indicating poor fitting performance. In contrast, the hybrid model (Figure 1D) achieves high comparability to the experimental response and the RSM, while displaying small SDs. Although RSM generates insights into the process at the endpoint, by solely considering this data point, it is not possible to find process optima, which are not located at the end of the process due to the limited evaluation.

The modeling of the soluble product titer proved to be more challenging. The RSM (Figure 2B) captures the overall shape but is not able to accurately represent the high values of the experimental response (Figure 2A). The black-box model (Figure 2C) performed similarly to the RSM. The prediction for replicate #4 in the test set almost matched the true response, while the test set replicate #22 was still displayed inaccurately. The hybrid model (Figure 2D) was able to predict these replicates more accurately and, overall, displayed the highest similarity to the experimental response while maintaining small SDs.

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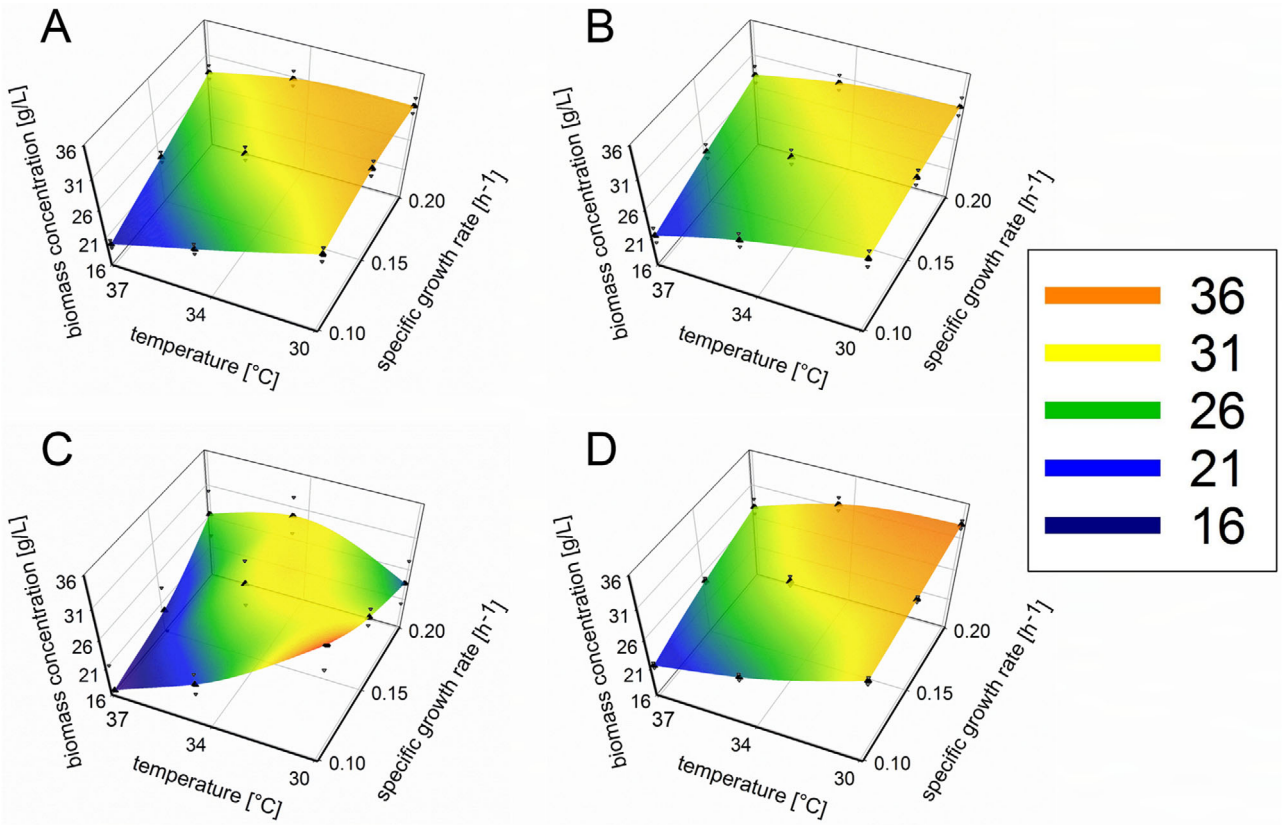

Figure 1. Response surface predictions of the endpoint values of the biomass concentration. A) The experimental response, B) the full quadratic RSM, C) the bootstrap-aggregated black-box model, and D) the bootstrap-aggregated hybrid model are displayed as a function of the temperature and the intended specific growth rate for the induction plane 0.5 μmol IPTG/g CDM. The color indicates the values of the biomass concentration from dark blue (lowest value) to orange (highest value). For the analytical measurements and the values derived from the respective model (triangles) the SD is indicated.

3.2. Time-Resolved Model Performance Comparison

By the application of time-resolved models, the limitation of solely endpoint modeling and neglecting process dynamics, as is the case using RSM, can be overcome. To determine which time-resolved model is most suitable to accurately model the entire process, the predictions of the black-box model and the hybrid model were compared. The robustness and susceptibility to errors on the fed-batch fermentations from the test set were considered. The superiority of the hybrid model compared to a pure black-box model, with respect to predictability, is shown for the biomass concentration (**Figure 3**) and the soluble product titer (**Figure 4**). The complete comparison of the two model performances applied to the test set is given in Table S2, Supporting Information. The complete overview of the scatter and the time-resolved plots for the training and the test set, including the error bars of the prediction (deliberately omitted here), as was generated by the Novasign hybrid modeling toolbox, is shown in Figures S2 (ANN black box) and S3 (hybrid model), Supporting Information.

Considering the biomass concentration, a high spreading of the values in both, the training and test partition, for the bootstrap-aggregated ANN black-box model (Figure 3A) was observed, while the bootstrap-aggregated hybrid model displayed a tight distribution around the least-squares line (Figure 3B). The predictive error for the biomass concentration decreased threefold due to the incorporated knowledge in the structure of the hybrid model. Similar results were obtained when considering the time-resolved model performance on the test set. The predicted values from the bootstrap-aggregated ANN black-box model clearly differed from the analytical measurements, resulting in broad PIs. This was even observed for the prediction of the replicates (Figure 3C,D), indicating an incapacity to fit the data.

The bootstrap-aggregated hybrid model did not display such issues and predicted the trend correct with small PIs. Likewise, this was the case for the fed-batch fermentations, which were not present in the training set. The biomass predictions of the bootstrap-aggregated ANN black-box model (Figure 3E) did not match the analytical results, being even outside the PIs. The bootstrap-aggregated hybrid model was able to predict these trends accurately. Only for one fed-batch fermentation, also the bootstrap-aggregated hybrid model failed to match the analytical values, still predicting the correct trend but in an insufficient manner, while the bootstrap-aggregated ANN black-box model again displayed fluctuating values (Figure 3F).

For the soluble product titer, the results of the scatter plots for the bootstrap-aggregated ANN black-box model (Figure 4A) and

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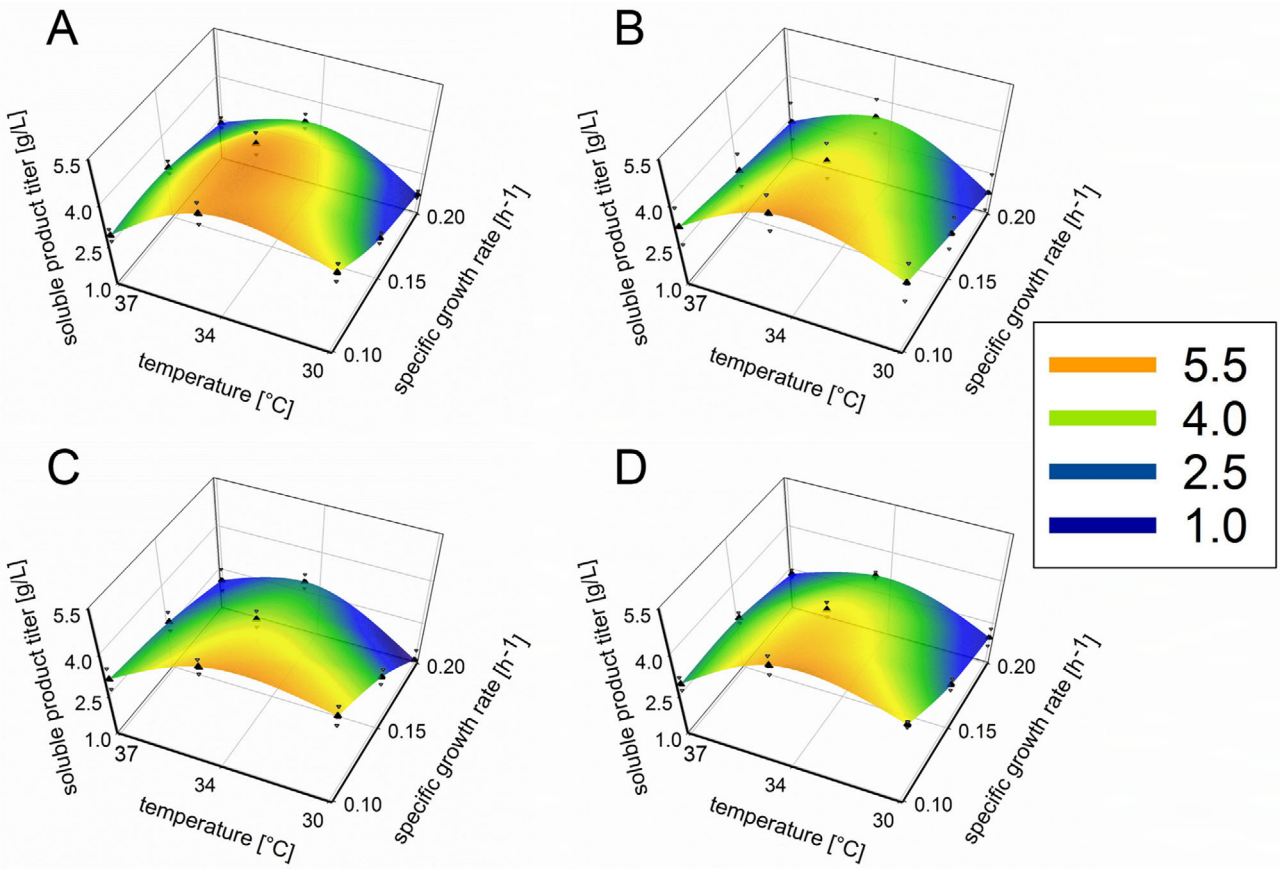

Figure 2. Response surface predictions of the endpoint values of the soluble product titer. A) The experimental response, B) the full quadratic RSM, C) the bootstrap-aggregated black-box model, and D) the bootstrap-aggregated hybrid model are displayed as a function of the temperature and the intended specific growth rate for the induction plane 0.5 µmol IPTG/g CDM. The color indicates the values of the soluble product titer from dark blue (lowest value) to orange (highest value). For the analytical measurements and the values derived from the respective model (triangles) the SD is indicated.

Table 1. Performance results of the RSM, the bootstrap-aggregated black-box model, and the bootstrap-aggregated hybrid model predicting the endpoint values of the biomass concentration and the soluble product titer. For each process variable, the model, the RMSE, and the percentage error are given (rounded to two decimal places). Additional model characteristics are indicated. The number of fed-batch fermentations used for model training and model testing is indicated in brackets.

Endpoint values	Biomass			Product		
Performance criteria	state of the art RSM ($n = 31$)	ANN black-box model ($n = 25+6$)	Hybrid model ($n = 25 + 6$)	state of the art RSM ($n = 31$)	ANN black-box model ($n = 25 + 6$)	Hybrid model ($n = 25 + 6$)
RMSE [g L^{-1}]	0.95	4.32	0.99	0.47	0.47	0.33
Error [%]	2.84	11.62	3.06	21.45	19.11	9.54
Entire process modeling	×	✓	✓	×	✓	✓
Captures process dynamics	×	✓	✓	×	✓	✓
Incorporation of process knowledge	×	×	✓	×	×	✓

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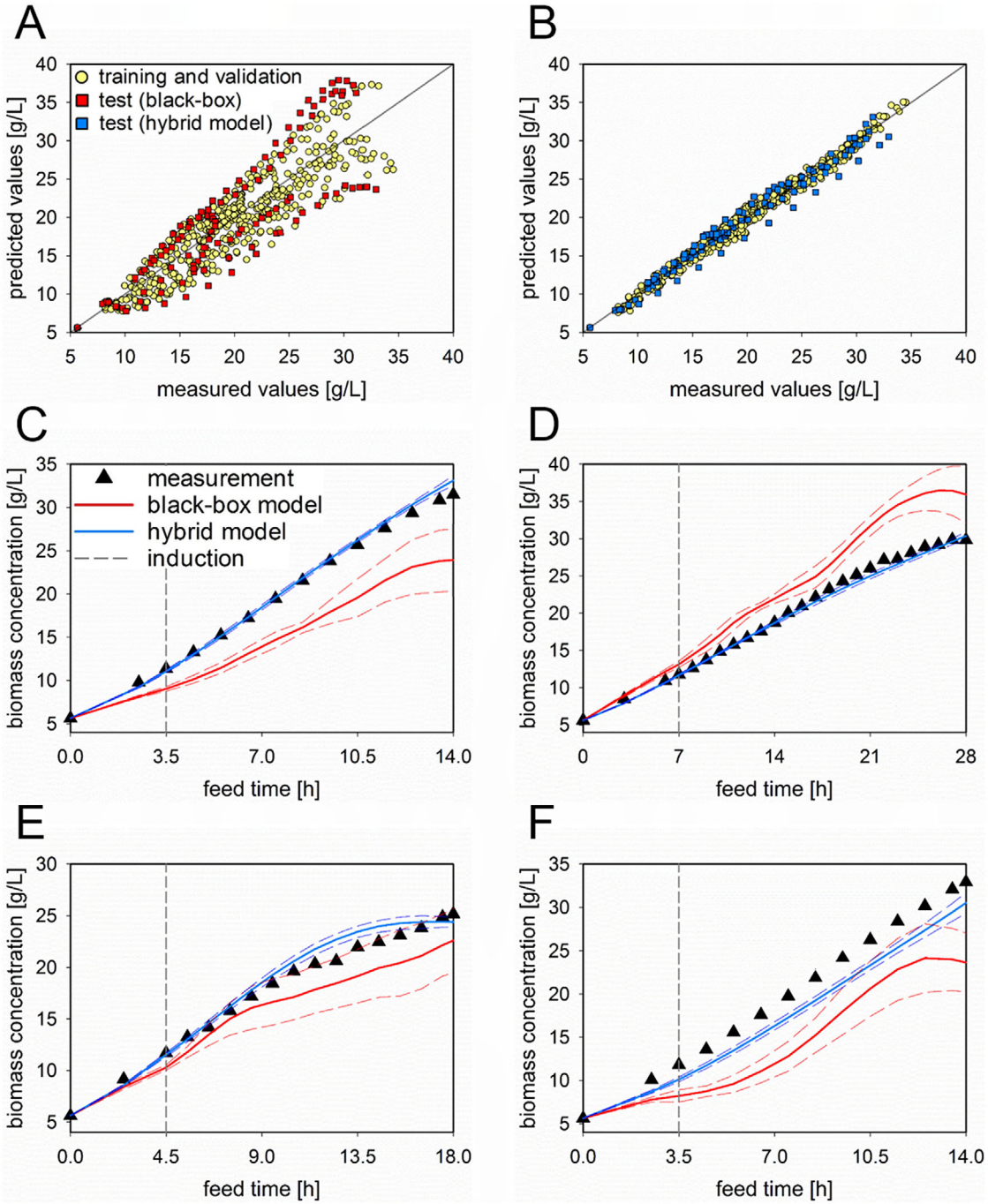

Figure 3. Performance comparison of the bootstrap-aggregated ANN black-box and the bootstrap-aggregated hybrid modeling approaches applied to the test set. The analytical results for the biomass concentration and the predictions from the two models are presented. A) The scatter plot of the ANN black-box model, B) the hybrid model and the time-resolved comparison of four fed-batch fermentations of the test set are displayed, including the respective PI of the model. C) DoE #22, D) DoE #4, E) DoE #17, and F) DoE #21.

the bootstrap-aggregated hybrid model (Figure 4B) did not differ as much as for the biomass. Still, the predictions from the test set of the bootstrap-aggregated hybrid model presented narrower distributions compared to the bootstrap-aggregated ANN black box. Likewise as for the biomass concentrations, the bootstrap-aggregated hybrid model was able to predict the soluble product titer trends of the fed-batch fermentations accurately. The predictions obtained from the bootstrap-aggregated ANN black-box model did not match the analytical measurements of the replicate in Figure 4C, but performed sufficiently on the second replicate (Figure 4D). The predicted trends of the fed-batch fermentation in Figure 4E was slightly underestimated by the

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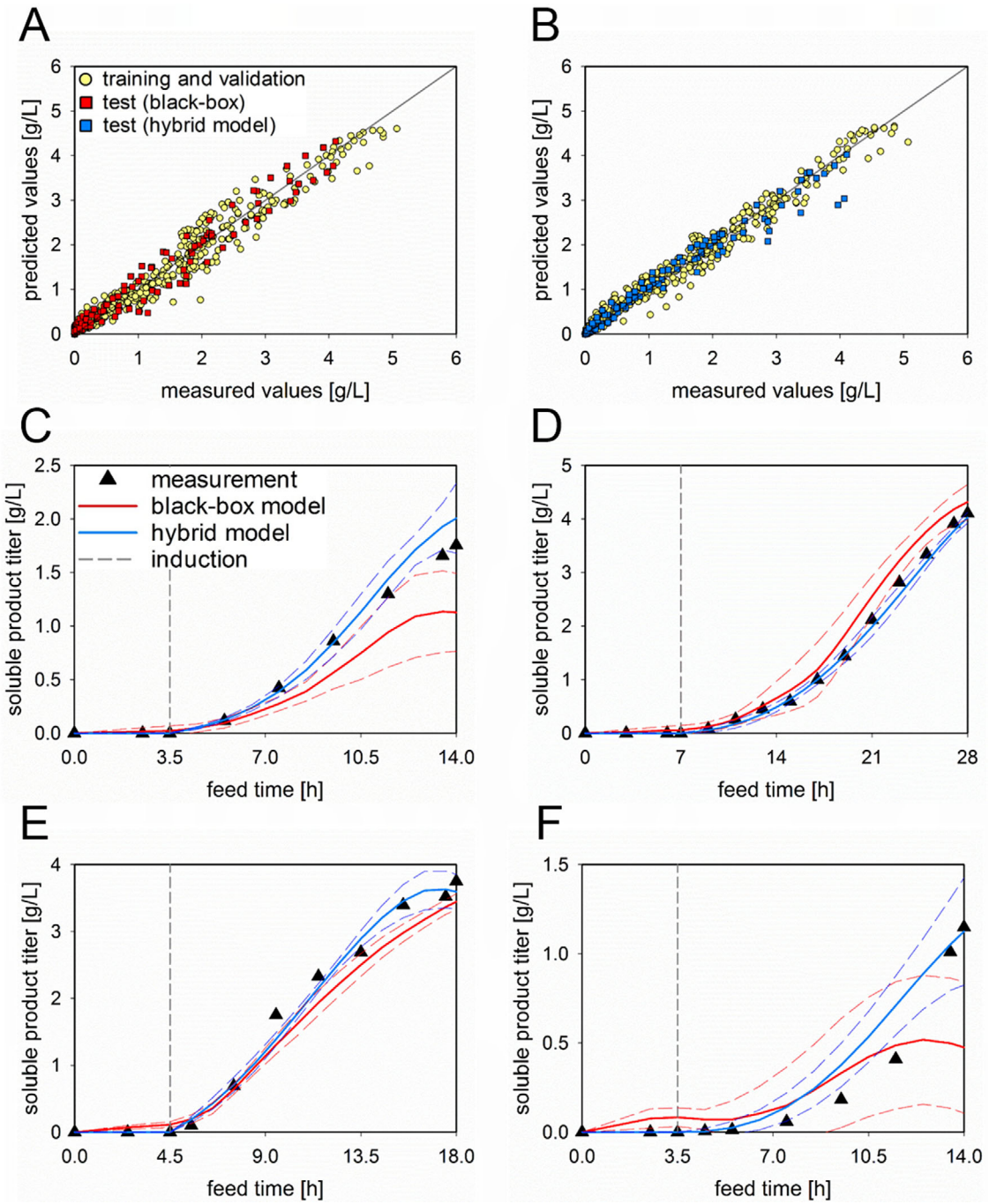

Figure 4. Performance comparison of the bootstrap-aggregated ANN black-box and the bootstrap-aggregated hybrid modeling approaches applied to the test set. The analytical results for the soluble product titer and the predictions from the two models are presented. A) The scatter plot of the ANN black-box model, B) the hybrid model and the time-resolved comparison of four fed-batch fermentations of the test set are displayed, including the respective PI of the model. C) DoE #22, D) DoE #4, E) DoE #17, and F) DoE #21.

bootstrap-aggregated ANN black-box model, but well described by the bootstrap-aggregated hybrid model. The only fed-batch fermentation in which the bootstrap-aggregated hybrid model did not perform well, was the same as for the biomass concentration before and also the bootstrap-aggregated ANN black-box model displayed the highest deviation from the analytical measurements in this cultivation (Figure 4F). Both bootstrap-aggregated models displayed the broadest PIs. To find possible explanations for the prediction difficulty of the fed-batch fermentation in Figures 3F and 4F, the respective rates, calculated from the off-line measurements, which are estimated in the black box of the hybrid model structure, of all fed-batch fermentations from the test set were investigated and are shown in Figure S4, Supporting Information.

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Moreover, all time-resolved depictions of the bootstrap-aggregated ANN black-box model already predict product before induction. Due to the inductor switch in the white box of the hybrid model, such product overestimations before induction were eliminated. Moreover, the chosen CPPs for the ANN model input may be suboptimal for predicting the variables of interest. Therefore, these might not be the most suitable factors to completely describe the behavior in the characterized space. The predictability of the soluble product titer increased, reducing the range of the PIs, applying the bootstrap-aggregated hybrid model since herein the biomass which can be predicted with high accuracy is linked to the product.

4. Discussion

It has been shown that the widely adopted RSM is able to fit the endpoint values of the characterized space quite accurately (Figures 1B and 2B). This fitting function contained constant, linear, and squared terms and interactions between the CPPs. However, typically only one point in time, mostly the endpoints, are used for model development, disregarding completely the evolution of the process over time. Herein, several drawbacks are recognizable, when process parameters are assumed to be constant during the process. For instance, the impact of temporal deviations in the CPPs on the endpoint response cannot be assessed. Also the desired process optimum, for example, the highest specific concentration may be overlooked. In contrast, the developed ANN black-box model and the hybrid model enable modeling of the biomass concentration and the soluble product titer of the process in a time-resolved manner. Moreover, by the utilization of bootstrapping the uncertainty of the developed model is visible.

When taking only the endpoint values into account, the superiority of the hybrid model over the black-box model could be determined (Figures 1C,D and 2C,D). Also, the hybrid model performed comparatively well to the RSM and displayed the smallest SD, while providing the additional advantage of modeling the entire process. Therefore, being able to identify process optima during the whole process. The same tendency was observed for the soluble product titer, for which in general less accurate prediction results were obtained.

Most probably, this behavior can be explained by the combination of various factors. The most prominent issue could have been a non-producing subpopulation without plasmids which increases during the cultivations but was not measured analytically. Especially for the cultivations at low feeding rates with high induction and high temperatures (DoE #9) this behavior can be seen in the increased growth rates at the end of the cultivation (Figure S4C, Supporting Information) Therefore, the modeling could not account for it. Another point to consider was the formation of inclusion bodies at certain CPP combinations, which decreased the soluble product. The initial variability of the plasmid copy number and the occurring cell lysis were also considered as influencing factors, which could not be taken into account. Addressing total SOD production did also not yield more accurate results due to the higher uncertainty of inclusion body analytics via SDS gel electrophoresis.

With respect to modeling the entire process, it also has been demonstrated that the bootstrap-aggregated hybrid model outperforms the bootstrap-aggregated black-box model (Figures 3 and 4). Due to the hybrid model structure, the model gains the advantage of understanding whether occurring variations are due to changes in the metabolism (e.g., the exemplary rate expressions in Figure S4, Supporting Information) or process operations, for example, the feeding, resulting in small SDs and tight PIs. The bootstrap-aggregated black-box model does not possess this ability; resulting in imprecise predictions for the process variable trajectories. This limitation was not observed using the bootstrap-aggregated hybrid model, due to the corrective action of the white box.

These results are showing that hybrid modeling has become a reliable and highly beneficial concept for upstream process characterization, including better process characterization and building a dynamic model. The developed bootstrap-aggregated hybrid model keeps up with commonly used techniques predicting the endpoint values of the process and even outperforms these techniques with respect to time-resolved process modeling. Moreover, the three CPPs used as inputs to the hybrid model are controllable, which in principle enables MPC in future applications.

Even though we could not account for every limitation in the presented data set, as mentioned above, the bootstrap-aggregated hybrid model performed reliably and with high predictability. To gain superior results in possible future DoE studies, more analytical techniques should be applied to also consider the above-mentioned factors. This includes on-line measurements such as off-gas analysis (CO_2 and O_2) and off-line analysis such as total organic carbon and measuring the DNA content in the suspension and supernatant to also access the carbon balance and to account for cell lysis.

In summary, to tackle these previously mentioned research gaps, regarding process understanding and QbD implementation, for example, in process characterization and optimization, the established bootstrap-aggregated hybrid model enables time-resolved modeling of the entire process for two target variables with a single model structure. This also renders possible the identification of process optima differing from the endpoint, potentially increasing the space time yield. We are aware of the fact that, while modeling the biomass concentration works well, our three chosen CPPs are probably insufficient inputs to ideally model the product formation for the ANN black box or the hybrid model. However, these CPPs were selected since they are easily controllable and predictable multisteps ahead, which is the prerequisite for subsequently building a model predictive or model-based control strategy. Therefore, these are promising control parameters to implement hybrid models for predictive soft sensors and also to implement MPC strategies, advancing from predictive to controllable models. While the first tool for advanced process control is already in use (a showcase of an exemplary fermentation, recorded in our OPC environment, is presented in Video S1, Supporting Information), the latter is currently in proof of concept and results will be presented in future publications.

An unsolved yet critical and major concern still is the long period of time spent on experiments until such a hybrid model is applicable. A novel approach to accelerate process characterization and optimization is presented by the concept of intensified DoE. Herein, by the utilization of intra-experimental shifts more than

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one CPP combination setting is addressed per cultivation run, reducing the total number of experiments.[40,41] This promising approach in combination with the herein developed bootstrap-aggregated hybrid modeling strategy will be investigated in future publications.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

Gerald Striedner and Mark Dürkop hold shares of Novasign GmbH.

Keywords

hybrid modeling, process control, Quality by Design

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